

FINITE POSITIVITY CERTIFICATES AND A CROSS-ENDPOINT OBSTRUCTION FOR A RIEMANN Ξ -DEFECT KERNEL

LEONARD VAN HEMERT

ABSTRACT. Let $\Xi(z) = \xi(\frac{1}{2} + iz)$ and let Φ denote the positive even Riemann theta kernel in its full-line Fourier representation. We study the two-variable cosine transform

$$C_y(x) = \iint_{\mathbb{R}^2} (u+v)^2 \operatorname{sinhc}(y(u+v)) \Phi(u) \Phi(v) \\ \times \cos(x(u-v)) \, du \, dv, \quad 0 \leq y \leq \frac{1}{2}.$$

The Hermite–Biehler defect of $E = \Xi + i\Xi'$ equals $2yC_y(x)$. Exact endpoint formulas, strict raw-moment monotonicity, and a global Taylor minorant for cosine lead to two results. First, outward-rounded Arb computation proves the zero-enumeration-free finite band

$$C_y(x) > 0 \quad \left(0 \leq y \leq \frac{1}{2}, \, |x| \leq R_{10}\right),$$

where

$$7.0362433 < R_{10} < 7.0362434.$$

Second, for every fixed $0 \leq \varepsilon < 1/2$, the initial positivity radii of the associated cross-endpoint polynomial hierarchy have finite supremum. The proof is analytic: the polynomial sections converge locally uniformly to a mixed trigonometric–hyperbolic transform that tends to $-\infty$ on the positive real axis. Thus this fixed endpoint-mismatch mechanism cannot yield unbounded positivity bands. This obstruction does not disprove the Riemann Hypothesis and does not apply to every moment or partition method.

1. INTRODUCTION

Write

$$(1) \quad \xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right).$$

The completed function satisfies

$$(2) \quad \xi(s) = \xi(1-s),$$

and Ξ is real entire and even. The Jacobi transformation gives the full-line representation

$$(3) \quad \Xi(z) = \int_{\mathbb{R}} \Phi(u) e^{izu} \, du,$$

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where, in the normalization used here,

$$(4) \quad \Phi(u) = \sum_{n=1}^{\infty} \left(4\pi^2 n^4 e^{9|u|/2} - 6\pi n^2 e^{5|u|/2} \right) e^{-\pi n^2 e^{2|u|}}.$$

For $u \geq 0$, each summand is positive because

$$4\pi^2 n^4 e^{9u/2} - 6\pi n^2 e^{5u/2} = 2\pi n^2 e^{5u/2} (2\pi n^2 e^{2u} - 3) > 0.$$

The Jacobi transformation makes the extension in (4) even and smooth. The kernel also satisfies a bound

$$(5) \quad 0 < \Phi(u) \leq C e^{9|u|/2} \exp\left(-\frac{\pi}{2} e^{2|u|}\right)$$

for a fixed $C > 0$.

Moment inequalities, Fourier criteria, and Hermite–Biehler formulations for Ξ have a substantial history. Csordas, Norfolk, and Varga proved Pólya–Turán inequalities for Taylor coefficients [3]. Csordas and Varga proved strict concavity of $t \mapsto \log \Phi(\sqrt{t})$ and obtained broad moment inequalities [4]. Their later work developed Hermite–Biehler and generalized Laguerre criteria [5, 6]. Coffey and Csordas gave another proof of logarithmic concavity and related moment inequalities [2], while Csordas studied positive-definite associated kernels and isolated global positivity conditions equivalent to the Riemann Hypothesis [1].

The first contribution here combines moments at $y = 0$ and $y = \frac{1}{2}$ to obtain one explicit polynomial lower bound for a two-variable defect kernel, uniformly over the closed y -interval. The endpoint moments are finite combinations of derivatives of ξ at 0 and $\frac{1}{2}$, so Arb can certify them without locating zeros. The first positive root gives the finite band.

The second contribution determines the fate of the same construction at all orders. For each fixed endpoint mismatch $0 \leq \varepsilon < 1/2$, the lower polynomials converge to an explicit mixed transform. The hyperbolic part coming from $y = 1/2$ has a strictly larger coefficient than the part coming from $y = \varepsilon$, forcing the limit to $-\infty$. Consequently the certified radii of this hierarchy are bounded. The conclusion is a no-go theorem for this specific lower-polynomial mechanism, not for the Riemann Hypothesis or for endpoint moments in general.

The zero consequence is not a record. Platt and Trudgian rigorously verified the Riemann Hypothesis through height $3 \cdot 10^{12}$ [7]. The interest of the present result lies in the moment certificate and its uniform kernel inequality, rather than the small numerical height of the corollary.

2. THE DEFECT KERNEL

Set

$$\operatorname{sinhc} t = \begin{cases} \sinh(t)/t, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

For $p = u + v$ and $q = u - v$, define

$$(6) \quad W_y(u, v) = p^2 \sinhc(y p) \Phi(u) \Phi(v), \quad 0 \leq y \leq \frac{1}{2},$$

and

$$(7) \quad C_y(x) = \iint_{\mathbb{R}^2} W_y(u, v) \cos(xq) \, du \, dv.$$

For $y > 0$, this equals the form

$$C_y(x) = \iint_{\mathbb{R}^2} \frac{p \sinhc(y p)}{y} \Phi(u) \Phi(v) \cos(xq) \, du \, dv.$$

Estimate (5) shows that

$$|p|^a |q|^b e^{p|/2} \Phi(u) \Phi(v)$$

is integrable for every fixed pair of nonnegative integers a, b . Thus all integrals and derivatives used below converge absolutely and locally uniformly. Fubini's theorem, differentiation under the integral, and the limits at $y = 0$ and $y = \frac{1}{2}$ are valid.

Lemma 2.1 (Defect identity). *Let*

$$E(z) = \Xi(z) + i\Xi'(z), \quad E^\#(z) = \overline{E(\bar{z})}.$$

Then

$$(8) \quad |E(x + iy)|^2 - |E^\#(x + iy)|^2 = 2yC_y(x)$$

for real x and $0 \leq y \leq \frac{1}{2}$.

Proof. Put $s = \frac{1}{2} - y$ and

$$P(s, x) = \xi(s + ix)\xi(s - ix) = |\Xi(x + iy)|^2.$$

Multiply the Fourier integrals. Symmetrization under $u \leftrightarrow v$ and under simultaneous sign reversal of u and v gives

$$(9) \quad P(s, x) = \iint_{\mathbb{R}^2} \Phi(u) \Phi(v) \coshc(y p) \cos(xq) \, du \, dv.$$

Consequently,

$$(10) \quad \frac{\partial P}{\partial y}(s, x) = \iint p \sinhc(y p) \Phi(u) \Phi(v) \cos(xq) \, du \, dv = yC_y(x).$$

Because Ξ is real entire, $E^\# = \Xi - i\Xi'$. Direct expansion yields

$$|E|^2 - |E^\#|^2 = -4 \operatorname{Im}(\Xi' \bar{\Xi}) = 2 \frac{\partial}{\partial y} |\Xi(x + iy)|^2.$$

Equation (10) proves (8). □

At $y = 0$, differentiation of (10) gives

$$(11) \quad C_0(x) = 2 \left(\Xi'(x)^2 - \Xi(x) \Xi''(x) \right).$$

3. ENDPOINT MOMENTS

For $j \geq 0$, define the raw moments

$$(12) \quad A_{2j}(y) = \iint_{\mathbb{R}^2} W_y(u, v) q^{2j} du dv.$$

Then

$$(13) \quad C_y^{(2j)}(0) = (-1)^j A_{2j}(y),$$

and $A_{2j}(y) > 0$.

Define

$$(14) \quad H_{2j}(s) = \sum_{\ell=0}^{2j} (-1)^\ell \binom{2j}{\ell} \xi^{(\ell)}(s) \xi^{(2j-\ell)}(s).$$

Lemma 3.1 (Moment identities). *For $y > 0$,*

$$(15) \quad A_{2j}(y) = -\frac{H'_{2j}(\frac{1}{2} - y)}{y}.$$

The continuous endpoint values are

$$(16) \quad A_{2j}(0) = H''_{2j}\left(\frac{1}{2}\right), \quad A_{2j}\left(\frac{1}{2}\right) = -2H'_{2j}(0).$$

Proof. Analytic continuation of (3) gives

$$\xi(s) = \int_{\mathbb{R}} \Phi(u) e^{(s-1/2)u} du.$$

Expanding $(u - v)^{2j}$ proves

$$H_{2j}(s) = \iint_{\mathbb{R}^2} e^{(s-1/2)p} q^{2j} \Phi(u) \Phi(v) du dv.$$

Equivalently, Leibniz differentiation gives

$$\partial_x^{2j} P(s, x) \Big|_{x=0} = (-1)^j H_{2j}(s).$$

Since $C_y = -P_s/y$, equations (13) and (15) follow. The functional equation implies

$$H_{2j}(1 - s) = H_{2j}(s), \quad H'_{2j}(1/2) = 0.$$

Taylor expansion at $1/2$ gives the first endpoint formula; substituting $y = 1/2$ gives the second. $\square$

Lemma 3.2 (Strict raw-moment monotonicity). *For every $j \geq 0$,*

$$0 \leq y_1 < y_2 \leq \frac{1}{2} \implies A_{2j}(y_1) < A_{2j}(y_2).$$

Consequently, with $m_{2j}(y) = A_{2j}(y)/A_0(y)$,

$$(17) \quad \frac{A_{2j}(0)}{A_0(1/2)} < m_{2j}(y) < \frac{A_{2j}(1/2)}{A_0(0)}$$

for every $y \in [0, 1/2]$ and $j \geq 1$.

Proof. For $p \neq 0$,

$$\sinhc(y p) = \sum_{r=0}^{\infty} \frac{y^{2r} p^{2r}}{(2r+1)!}$$

is strictly increasing for $y \geq 0$. The remaining factor in (12) is nonnegative and is positive away from measure-zero lines. This proves strict monotonicity. Applying the endpoint bounds separately to the numerator and denominator gives (17). Equality would require the numerator and denominator to attain opposite endpoints at the same value of y , so both inequalities are strict even when y is an endpoint. $\square$

4. THE DEGREE-TEN LOWER POLYNOMIAL

Lemma 4.1 (Global cosine minorant). *For every real $r \neq 0$,*

$$(18) \quad \cos r > T_{10}(r) := 1 - \frac{r^2}{2!} + \frac{r^4}{4!} - \frac{r^6}{6!} + \frac{r^8}{8!} - \frac{r^{10}}{10!}.$$

Proof. For $g = \cos - T_{10}$,

$$g^{(10)}(r) = 1 - \cos r \geq 0,$$

and $g^{(k)}(0) = 0$ for $0 \leq k \leq 9$. Repeated integration proves $g(r) \geq 0$ for $r \geq 0$. The inequality is strict for $r > 0$ because $1 - \cos r$ does not vanish identically on any interval $(0, r)$. Evenness handles negative r . $\square$

For the five moments used below, set

$$(19) \quad \begin{aligned} B_2 &= \frac{A_2(1/2)}{A_0(0)}, & B_4 &= \frac{A_4(0)}{A_0(1/2)}, \\ B_6 &= \frac{A_6(1/2)}{A_0(0)}, & B_8 &= \frac{A_8(0)}{A_0(1/2)}, \\ B_{10} &= \frac{A_{10}(1/2)}{A_0(0)}. \end{aligned}$$

Define

$$(20) \quad P_{\theta}(x) = 1 - \frac{B_2 x^2}{2!} + \frac{B_4 x^4}{4!} - \frac{B_6 x^6}{6!} + \frac{B_8 x^8}{8!} - \frac{B_{10} x^{10}}{10!}.$$

Proposition 4.2 (Uniform polynomial lower bound). *For $0 \leq y \leq \frac{1}{2}$ and $x \neq 0$,*

$$(21) \quad \frac{C_y(x)}{A_0(y)} > P_{\theta}(x).$$

Proof. Normalize $W_y du dv$ by $A_0(y)$ and apply Lemma 4.1. Strictness holds because $xq \neq 0$ on a set of positive measure. We obtain

$$\frac{C_y(x)}{A_0(y)} > 1 - \frac{m_2(y)x^2}{2!} + \frac{m_4(y)x^4}{4!} - \frac{m_6(y)x^6}{6!} + \frac{m_8(y)x^8}{8!} - \frac{m_{10}(y)x^{10}}{10!}.$$

For negative terms, use the upper bounds in (17); for positive terms, use the lower bounds. This gives (21). $\square$

5. CERTIFIED ENDPOINT ARITHMETIC

The certificate evaluates (16) from Taylor balls for

$$(22) \quad \xi(s) = (s-1)\pi^{-s/2}\Gamma(1+s/2)\zeta(s),$$

which is algebraically identical to (1) and analytic at both centers 0 and 1/2.

Table 1 lists ball centers. At 269-bit Arb precision, the largest radius among the twelve displayed values is $1.37 \cdot 10^{-67}$.

TABLE 1. Certified endpoint-moment ball centers.

Moment	$A_{2j}(0)$	$A_{2j}(1/2)$
A_0	0.02283966166888743667469247	0.02309570896612103381431025
A_2	0.001890366586781360982949785	0.001909616896683607096775872
A_4	0.000459720156209605774158694	0.000464013763611390134012507
A_6	0.000182390894103604245533460	0.000183966337058884295480848
A_8	0.000099156363571899705593806	0.000099954209940825929544209
A_{10}	0.000067852553023503413540728	0.000068364291172414501807343

The resulting coefficient balls have centers

$$(23) \quad \begin{aligned} B_2 &= 0.0836096840823574320891615416 \dots, \\ B_4 &= 0.0199050029979147510756695686 \dots, \\ B_6 &= 0.00805468748731450212309766654 \dots, \\ B_8 &= 0.00429328078723851344898443484 \dots, \\ B_{10} &= 0.00299322696472082444610981628 \dots \end{aligned}$$

Put $Q(t) = P_\theta(\sqrt{t})$. On $0 \leq t \leq 51$, the degree-four Bernstein coefficient balls for Q' have centers

$$(24) \quad \begin{aligned} &-0.0418048420411787160, \quad -0.0206557763558942930, \\ &-0.0140554899445716895, \quad -0.00787928068151705499, \\ &-0.0159039318562651082. \end{aligned}$$

Every complete ball is negative. Since Bernstein basis functions are nonnegative on the unit interval, $Q'(t) < 0$ for $0 \leq t \leq 51$.

For $t = 51 + r$, write

$$Q'(51 + r) = \sum_{j=0}^4 c_j r^j.$$

The coefficient-ball centers are

$$(25) \quad \begin{aligned} c_0 &= -0.0159039318562651082, \\ c_1 &= -0.000629384405862592411, \\ c_2 &= -0.0000327586169268804792, \\ c_3 &= -0.000000415429458942178156, \\ c_4 &= -0.000000004124265548832706. \end{aligned}$$

Again every complete ball is negative, so $Q'(t) < 0$ for $t \geq 51$. Hence

$$(26) \quad Q'(t) < 0 \quad (t \geq 0).$$

The exact rational evaluation points satisfy the certified inequalities

$$(27) \quad P_\theta(7.0362433) > 3.7223537277 \cdot 10^{-9} > 0,$$

$$(28) \quad P_\theta(7.0362434) < -1.7438217651 \cdot 10^{-8} < 0.$$

Because $Q(0) = 1$, (26) holds, and the leading coefficient of Q is negative, P_θ has exactly one positive zero. Denote it by R_{10} . Then

$$(29) \quad 7.0362433 < R_{10} < 7.0362434, \quad R_{10} = 7.03624331759098920168 \dots$$

Theorem 5.1 (Certified finite positivity band). *For every $0 \leq y \leq \frac{1}{2}$,*

$$(30) \quad C_y(x) > 0 \quad (|x| \leq R_{10}).$$

Proof. For $0 < |x| < R_{10}$, Proposition 4.2 and $P_\theta(x) > 0$ prove the claim. At $|x| = R_{10}$, strictness in (21) gives

$$\frac{C_y(R_{10})}{A_0(y)} > P_\theta(R_{10}) = 0.$$

At $x = 0$, $C_y(0) = A_0(y) > 0$. Evenness in x completes the proof. $\square$

6. FINITE ZERO CONSEQUENCE

Corollary 6.1. *If ρ is a nontrivial zero of ζ with $|\operatorname{Im} \rho| \leq R_{10}$, then*

$$\operatorname{Re} \rho = \frac{1}{2}.$$

Proof. At a zero of Ξ , the two functions E and $E^\#$ have the same modulus, so the left side of (8) vanishes. If $\rho = \beta + i\gamma$ with $\beta < 1/2$, then

$$z = \gamma + i(1/2 - \beta)$$

satisfies $\Xi(z) = \xi(\rho) = 0$ and $0 < \operatorname{Im} z < 1/2$. For $|\gamma| \leq R_{10}$, Theorem 5.1 and (8) make the defect positive, a contradiction. If $\beta > 1/2$, the functional equation and conjugation give the zero $1 - \bar{\rho}$, with the same ordinate and real part below $1/2$. $\square$

At a real zero x of Ξ , equation (11) becomes $C_0(x) = 2\Xi'(x)^2$. Thus Theorem 5.1 also implies simplicity of any real Ξ -zero in the finite interval, without assuming simplicity.

7. FAILURE FOR A GENERIC POSITIVE KERNEL

The proof uses positivity and decay in its analytic steps, but the numerical moment profile is specific to the Riemann kernel. Consider

$$(31) \quad \Phi_\varepsilon(u) = \psi_\varepsilon(u-1) + \psi_\varepsilon(u+1) + \frac{13}{6}\psi_\varepsilon(u), \quad \psi_\varepsilon(u) = e^{-2 \cosh(u/\varepsilon)}.$$

This kernel is smooth, positive, even, and double-exponentially decreasing. Its Fourier transform factors as

$$(32) \quad \widehat{\Phi}_\varepsilon(z) = 2\varepsilon K_{i\varepsilon z}(2) \left(2 \cos z + \frac{13}{6} \right).$$

With $\lambda = \log(3/2)$, one has $2 \cosh \lambda = 13/6$, so

$$z = (2k+1)\pi \pm i\lambda$$

are exact nonreal zeros and $0 < \lambda < 1/2$.

The failure appears already in the normalized moments. Let $a = 13/6$ and let $m_{2j}^{(\varepsilon)}(y)$ denote the moments formed from (31). Scaling each of the three bumps and applying dominated convergence gives, for $j \geq 1$,

$$\frac{A_{2j}^{(\varepsilon)}(y)}{\varepsilon^2 M^2} \longrightarrow 4a \operatorname{sinhc} y, \quad M = \int_{\mathbb{R}} e^{-2 \cosh t} dt,$$

while

$$\frac{A_0^{(\varepsilon)}(y)}{\varepsilon^2 M^2} \longrightarrow 8 \operatorname{sinhc}(2y) + 4a \operatorname{sinhc} y.$$

Therefore

$$(33) \quad \lim_{\varepsilon \rightarrow 0} m_{2j}^{(\varepsilon)}(y) = \frac{a}{2 \cosh y + a}.$$

At $y = \lambda$, this limit is $1/2$, which violates the Riemann-kernel bound $m_2(y) < B_2 = 0.0836096840\dots$ for all small enough ε . At $y = 0$, the limit is $13/25$.

For $0 < \varepsilon \leq 1/4$, the same family also satisfies

$$\Phi_\varepsilon(1/2)^2 < \Phi_\varepsilon(0)\Phi_\varepsilon(1),$$

so it is not log-concave. Neither positivity nor rapid decay supplies the endpoint ratios in (19).

8. THE CROSS-ENDPOINT HIERARCHY

The degree-ten theorem does not control the large-frequency sign of C_y . The exact Taylor hierarchy makes the obstruction precise. For $m \geq 0$ set

$$(34) \quad S_m(y, x) = \sum_{j=0}^{2m+1} (-1)^j \frac{m_{2j}(y)x^{2j}}{(2j)!}.$$

The global Taylor inequality $T_{4m+2}(r) < \cos r$ for $r \neq 0$ gives

$$(35) \quad S_m(y, x) < \frac{C_y(x)}{A_0(y)} \quad (x \neq 0).$$

Proposition 8.1 (Exact hierarchy). *Fix $\varepsilon > 0$ and $R < \infty$. Then S_m converges uniformly to $C_y(x)/A_0(y)$ on*

$$[\varepsilon, 1/2] \times [-R, R].$$

Moreover, $C_y(x) > 0$ on this rectangle if and only if some S_m is uniformly positive there.

Proof. Estimate (5) gives a uniform exponential moment bound

$$\sup_{\varepsilon \leq y \leq 1/2} \int e^{R|q|} \frac{W_y(u, v)}{A_0(y)} du dv < \infty.$$

Dominated convergence for the cosine series gives uniform convergence. If C_y/A_0 is positive on the compact rectangle, it has a positive minimum; uniform convergence makes every sufficiently high section positive. The reverse implication follows from (35), with the value at $x = 0$ equal to 1. $\square$

Thus divergence of the positivity radii for the exact sections is equivalent to global open-strip positivity and supplies no weaker intermediate theorem. The cross-endpoint hierarchy is more restrictive. For $\varepsilon \geq 0$, define

$$L_{2j}^{(\varepsilon)} = \frac{A_{2j}(\varepsilon)}{A_0(1/2)}, \quad U_{2j}^{(\varepsilon)} = \frac{A_{2j}(1/2)}{A_0(\varepsilon)},$$

and

$$(36) \quad \hat{P}_m^{(\varepsilon)}(x) = 1 + \sum_{\substack{1 \leq j \leq 2m+1 \\ j \text{ even}}} L_{2j}^{(\varepsilon)} \frac{x^{2j}}{(2j)!} - \sum_{\substack{1 \leq j \leq 2m+1 \\ j \text{ odd}}} U_{2j}^{(\varepsilon)} \frac{x^{2j}}{(2j)!}.$$

Then

$$\hat{P}_m^{(\varepsilon)}(x) < S_m(y, x) < \frac{C_y(x)}{A_0(y)}$$

for $x \neq 0$ and $\varepsilon \leq y \leq 1/2$. Theorem 5.1 is the case $m = 2$, $\varepsilon = 0$.

Let $\hat{R}_m^{(\varepsilon)}$ be the initial positivity radius of (36), namely the supremum of the initial interval on which it is positive. The next sections show that these radii cannot diverge for any fixed $\varepsilon < 1/2$.

9. THE EXACT MIXED LIMIT

For fixed $y \in [0, 1/2]$, define the hyperbolic transform

$$(37) \quad G_y(x) = C_y(ix) = \iint_{\mathbb{R}^2} W_y(u, v) \cosh(xq) du dv.$$

Although only real x is needed below, $C_y(z)$ is entire in z . Indeed, for every $R > 0$, estimate (5) gives

$$\iint_{\mathbb{R}^2} W_y(u, v) e^{R|q|} du dv < \infty$$

uniformly for $0 \leq y \leq 1/2$. Dominated convergence therefore gives the absolutely and locally uniformly convergent series

$$(38) \quad C_y(z) = \sum_{j=0}^{\infty} (-1)^j A_{2j}(y) \frac{z^{2j}}{(2j)!},$$

$$(39) \quad G_y(z) = \sum_{j=0}^{\infty} A_{2j}(y) \frac{z^{2j}}{(2j)!}.$$

Separating even and odd values of the index j yields

$$(40) \quad \sum_{j \text{ even}} A_{2j}(y) \frac{x^{2j}}{(2j)!} = \frac{G_y(x) + C_y(x)}{2},$$

$$(41) \quad \sum_{j \text{ odd}} A_{2j}(y) \frac{x^{2j}}{(2j)!} = \frac{G_y(x) - C_y(x)}{2}.$$

Fix $0 \leq \varepsilon < 1/2$ and abbreviate

$$(42) \quad a = A_0(\varepsilon), \quad b = A_0(1/2).$$

Strict raw-moment monotonicity gives $0 < a < b$.

Proposition 9.1 (Mixed-limit formula). *The polynomials $\hat{P}_m^{(\varepsilon)}$ converge locally uniformly on $\mathbb{C}$ to*

$$(43) \quad F_\varepsilon(x) = 1 - \frac{a}{b} + \frac{G_\varepsilon(x) + C_\varepsilon(x)}{2b} - \frac{G_{1/2}(x) - C_{1/2}(x)}{2a}.$$

Proof. Apply (40) to the positive even-index sum in (36) and (41) to its negative odd-index sum. The $j = 0$ term of the former is a/b , whereas the constant term of $\hat{P}_m^{(\varepsilon)}$ is defined to be one. This produces the essential correction $1 - a/b$. Local uniform convergence follows from (38)–(39). $\square$

10. FIXED ENDPOINT MISMATCH FORCES BOUNDED RADII

Lemma 10.1 (Hyperbolic growth). *For each fixed $0 \leq \varepsilon < 1/2$, there are $c, \delta > 0$ such that*

$$G_\varepsilon(x) \geq c \cosh(\delta x) \quad (x \geq 0).$$

In particular, $G_\varepsilon(x) \rightarrow +\infty$ as $x \rightarrow +\infty$.

Proof. The density W_ε is continuous and positive away from the null line $u + v = 0$. Choose a compact rectangle on which $W_\varepsilon > 0$ and $|u - v| \geq \delta$ for some $\delta > 0$. Its W_ε -mass c is positive. Restricting (37) to this rectangle gives the bound. $\square$

Theorem 10.2 (Cross-endpoint obstruction). *For every fixed $0 \leq \varepsilon < 1/2$,*

$$(44) \quad \sup_{m \geq 0} \hat{R}_m^{(\varepsilon)} < \infty.$$

This statement is unconditional and does not assume the Riemann Hypothesis.

Proof. For each fixed real p , the function $y \mapsto \sinhc(y p)$ is increasing on $[0, \infty)$. Therefore

$$W_{1/2}(u, v) \geq W_\varepsilon(u, v), \quad G_{1/2}(x) \geq G_\varepsilon(x) \quad (x \in \mathbb{R}).$$

Positivity of the measures also gives $|C_y(x)| \leq A_0(y)$. Formula (43) consequently implies

$$(45) \quad F_\varepsilon(x) \leq 1 - \frac{a}{b} + \frac{a}{2b} + \frac{b}{2a} - \frac{b-a}{2ab} G_\varepsilon(x).$$

The last coefficient is strictly negative because $a < b$, and Lemma 10.1 shows that the final term tends to $-\infty$. Hence $F_\varepsilon(x) \rightarrow -\infty$ as $x \rightarrow +\infty$.

Choose $X > 0$ with $F_\varepsilon(X) < 0$. Proposition 9.1 gives $\hat{P}_m^{(\varepsilon)}(X) < 0$ for every sufficiently large m , so $\hat{R}_m^{(\varepsilon)} \leq X$ for those m . Each remaining polynomial starts at one and has highest Taylor index $j = 2m + 1$, hence a strictly negative leading coefficient. It tends to $-\infty$ and has a finite first positive zero. Taking the maximum of X and the finitely many exceptional radii proves (44). $\square$

Remark 10.3 (Endpoint distinction). The proof includes $\varepsilon = 0$: $W_0 = p^2\Phi(u)\Phi(v)$ is the continuous endpoint weight and $A_0(0) > 0$. At $\varepsilon = 1/2$, however, $a = b$ and the coercive coefficient in (45) vanishes. Then (43) reduces to the exact normalized cosine transform. The obstruction theorem neither proves nor disproves its global positivity.

11. PARTITIONS AND THE RH-EQUIVALENCE BOUNDARY

On every fixed positive-width cell $[\alpha, \beta] \subset [0, 1/2]$, using positive coefficients from α and negative coefficients from β repeats the proof of Theorem 10.2. Thus a fixed finite partition improves finite constants but does not remove endpoint mismatch.

Mesh refinement is different. At fixed degree, continuity of the finitely many moments makes the cellwise bounds converge to the exact Taylor section. If degree and mesh are allowed to vary diagonally so that positivity is certified on every compact rectangle $[\varepsilon, 1/2] \times [-R, R]$, Proposition 8.1 yields global open-strip defect positivity. Through Lemma 2.1 this is the Hermite–Biehler condition equivalent to RH. Such a diagonal assertion is therefore not an established intermediate theorem; it reaches the original unresolved global condition.

Theorem 10.2 blocks the fixed cross-endpoint hierarchy only. It does not block exact moment sections. Nor does it block adaptive schemes with a new uniform estimate, or other approaches to RH.

12. HIGHER-DEGREE CERTIFIED COMPUTATION

The analytic proof above is independent of numerical experimentation. For reproducibility and finite-scale diagnosis, an independent Arb implementation nevertheless computed all 175 requested cases

$$m = 0, \dots, 20, 25, 30, 40, 50, \quad \varepsilon \in \{0, 1/4, 1/8, 1/16, 1/32, 10^{-2}, 10^{-3}\}.$$

It certified 816 endpoint moments through A_{202} , an exact dyadic bracket for every first positive root, and interval-Horner monotonicity up to that root. The successful full run used 700 decimal digits, a series cap of 205, 120 root-bisection bits, about 3.74 seconds wall time, and approximately 41.1 MB maximum resident memory. A 900-digit recomputation encloses every degree-202 result. The retained 180-digit run fails at $A_{120}(1/2)$ with an indeterminate ball; it is recorded rather than suppressed.

At $m = 50$, the extreme certified lower root endpoints are

$$7.5367678370859\dots \quad (\varepsilon = 0), \quad 7.7563175645352\dots \quad (\varepsilon = 1/4).$$

These are finite certificates, not the proof of boundedness.

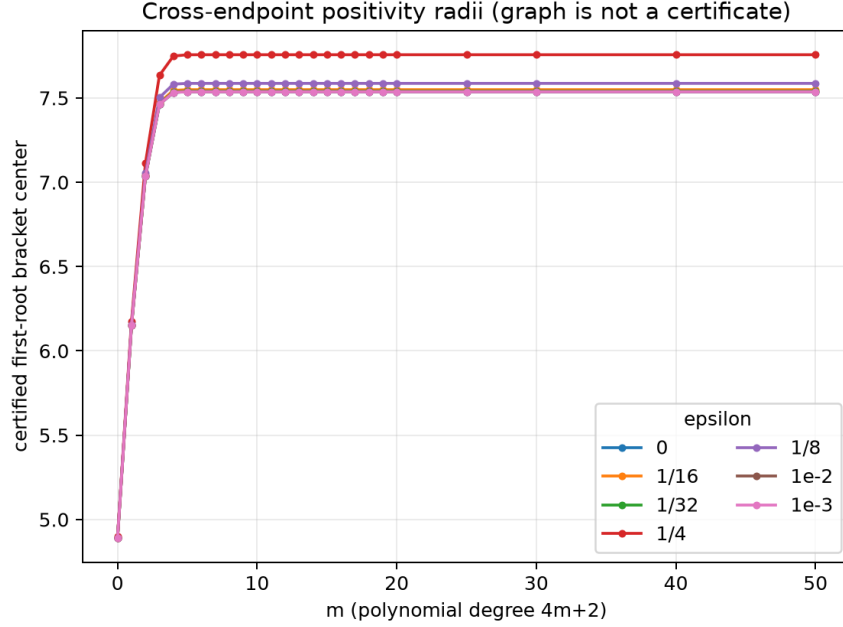


FIGURE 1. Certified initial positivity radii for the computed degrees and endpoint choices. The plotted finite data diagnose the hierarchy but do not establish an asymptotic law. Theorem 10.2 follows instead from the exact mixed limit.

The clean-room program independently reconstructs the degree-10 endpoint identities and root bracket. A second implementation checks derivative indexing, series caps, endpoint ratios, large-argument polynomial evaluation, and precision loss as $\varepsilon \rightarrow 0$. These are internal reproducibility and adversarial checks, not independent peer review.

13. REPRODUCIBILITY AND CERTIFICATE SCOPE

The public certificate evaluates Taylor balls for (22) at 0 and 1/2, forms (14), and extracts derivatives through order 12. It uses `python-flint` 0.9.0, FLINT 3.6.0, 269-bit Arb precision, and a formal-series cap of 18. The cap exceeds the thirteen coefficients required through degree 12.

The finite proof obligations are:

- (1) the twelve endpoint moment balls;
- (2) the five cross-endpoint ratios;
- (3) the five Bernstein signs in (24);

- (4) the five shifted-tail signs in (25);
- (5) the two rational root-bracket signs.

Arb implements midpoint-radius ball arithmetic with outward enclosures, so a strict comparison succeeds only when the entire output ball lies on the claimed side. The computation uses no quadrature and no list of zeta zeros. The analytic steps linking its finite values to C_y are Lemmas 2.1–4.1 and Proposition 4.2.

The repository records the exact dependency version, output, and SHA-256 checksums. The command is

```
uv run --with python-flint==0.9.0 \
  python code/theta_moment_certificate.py
```

Python optimization must remain disabled because program assertions encode the sign obligations. The higher-degree records and their independent checks are reproduced by

```
uv run --with python-flint==0.9.0 \
  python research/high_degree/verify_results.py
PYTHONPATH=research/adversarial \
  uv run --with python-flint==0.9.0 \
  python -m unittest -v \
  research/adversarial/test_adversarial.py
```

The dependency structure is:

- (1) the standard functional equation and Jacobi theta transformation give (3), positivity, and decay of Φ ;
- (2) those facts give absolute convergence, Lemmas 2.1–4.1, and the endpoint moments;
- (3) the finite Arb enclosures plus these analytic lemmas give Theorem 5.1;
- (4) decay alone gives the entire series and Proposition 9.1;
- (5) positivity, strict endpoint monotonicity, and the mixed limit give Theorem 10.2;
- (6) no numerical radius, zero list, RH assumption, zero-simplicity assumption, or saddle-point heuristic enters the obstruction theorem.

14. CONCLUSION

Endpoint derivatives of ξ at 0 and $1/2$, strict monotonicity of raw moments, and a global cosine minorant produce an explicit lower polynomial for C_y/A_0 . Arb certification of its coefficients and root proves the closed finite band in Theorem 5.1. At arbitrary degree, however, fixed cross-endpoint normalization creates an imbalanced hyperbolic term. The exact mixed limit tends to $-\infty$, and Theorem 10.2 proves that the resulting positivity radii remain bounded for every fixed $\varepsilon < 1/2$.

Thus the finite certificate is valid, while its fixed endpoint-mismatch extension cannot prove RH. This is not a theorem against RH, exact moment

sections, or every adaptive partition method. The remaining global open-strip positivity problem is unchanged and RH-equivalent.

DISCLOSURE AND LICENSING

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